Challenge 7: Rota's Basis Conjecture

The following conjecture in its original form [1, Conjecture 4] is stated in terms of matroids, but even in the special case where the matroid is a vector matroid – that is, the bases come from a finite-dimensional vector space – the conjecture is widely open.

Conjecture 1.1 (Rota's basis conjecture)**.** Let M be a finite matroid with a family $(B_1, .., B_n)$ of disjoint bases. There is a family of disjoint bases $(C_1, ..C_n)$ of M so that $B_i \cap C_j$ is a singleton for all $i, j \in [n]$.

Here we are interested in the following scaled weakening of the original conjecture.

Conjecture 1.2 (Scaled weakening of Rota's basis conjecture)**.** Let $r \in \mathbb{N}$. Let M be a finite matroid with a family $(B_1, .., B_n)$ of disjoint bases. There is $n_0 \in \mathbb{N}$ so that for all $n \geq n_0$ and all $\epsilon \in \mathbb{R}_{>0}$, there is a family of disjoint bases $(C_1, ..C_{\lceil (1-1/r-\epsilon)n \rceil})$ of M so that $B_i \cap C_j$ is a singleton for all $i, j \in [n]$.

Bucić, Kwan, Pokrovskiy and Sudakov proved the scaled weakening for $r = 2$ [2].

References

[1] R. Huang and G.-C. Rota. "On the relations of various conjectures on Latin squares and straightening coefficients". In: *Discrete Mathematics* 128.1-3 (1994), pp. 225–236.

[2] M. Bucić et al. "Halfway to Rota's basis conjecture". In: *International Mathematics Research Notices* 2020.21 (2020), pp. 8007–8026.